

SE Project
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Project: Webhook Ingestion & Retry Mechanism Hub

Architecture Selected: Microservices Architecture

Architectural Choice

We chose Microservices Architecture for the Webhook Ingestion & Retry Mechanism Hub. The system is divided into separate services for webhook ingestion, delivery, retry handling, logging, configuration, and dead-letter queue management. Each service performs a specific task and communicates with the other services through defined interfaces.

Reason 1 – Independent Scaling

The system may receive a large number of webhook requests, so some parts may need more resources than others. With microservices, the Webhook Ingestion Service and Delivery Service can be scaled separately depending on the traffic.

Reason 2 – Fault Isolation

If a target service is temporarily unavailable, the whole webhook system should not stop working. The Retry Manager can handle failed deliveries while the Ingestion Service continues receiving new webhook events.

Security Advantage

Different services can have separate access controls. Sensitive areas such as webhook logs, endpoint configuration, and retry management can be restricted to authorized users, while HTTPS/TLS can be used for webhook communication.

Performance Benefit

The Event Queue allows the system to accept webhook requests quickly without waiting for the target service to respond. Delivery can happen separately in the background, which helps the system handle higher traffic and prevents slow services from blocking new requests.